Food and Agriculture Organization of the United Nations

TECHNICAL NOTE

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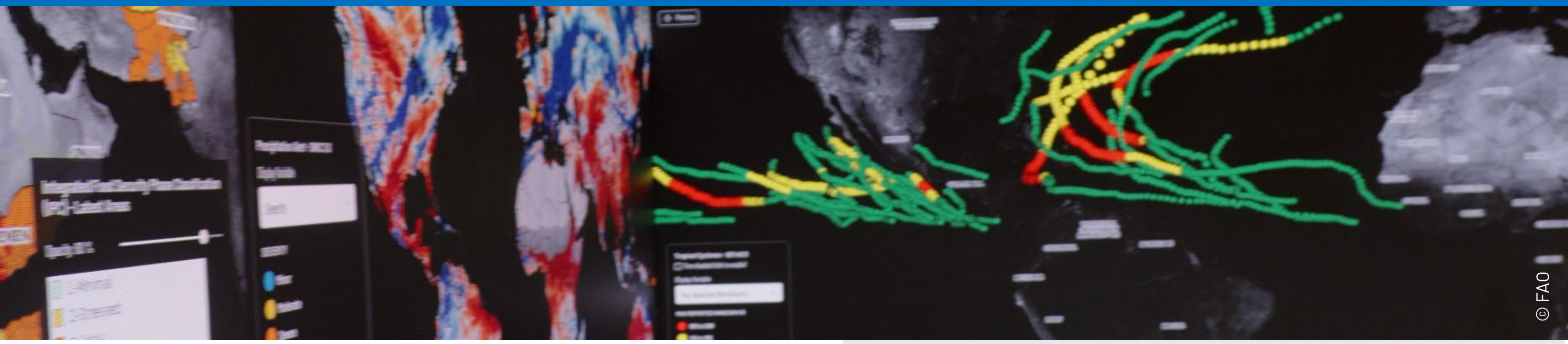


THE IMPACT OF OIL SUPPLY SHOCKS ON THE GLOBAL FOOD IMPORT BILL

Recent geopolitical events in the conflict in the Middle East have affected global energy markets, given the region's central role in oil production and its critical transit routes. Disruptions or threats to oil supply translate into higher international oil prices, which act as a global cost shock for agrifood systems.

Oil plays an essential role in food production, processing, transportation, and cold storage. Consequently, disruptions in oil supply affect food prices and increase food import bills (FIB), which is a major concern for nations that rely on imported food.

The impact of an oil supply shock can be compounded by elevated geopolitical risks, which raises concerns that oil shocks may have non-linear and amplified effects compared to more tranquil periods. This reflects heightened uncertainty, precautionary behaviour, tighter financial conditions, and stronger pass-through of energy costs along global supply chains during episodes of increased geopolitical stress.

KEY MESSAGES

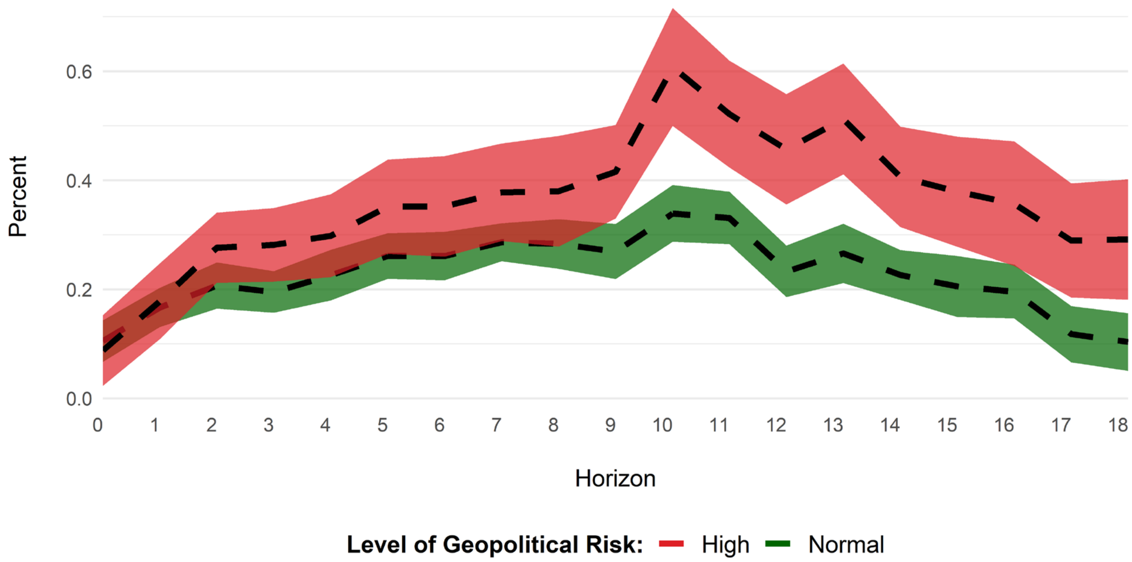
- Oil supply shocks raise global food import bills, reflecting the central role of energy in food production, processing, and transportation. Even moderate oil price increases translate into higher food import costs, especially for net food importing countries.
- Geopolitical risk strongly amplifies these effects. When oil supply shocks occur during periods of elevated geopolitical tensions, the impact on the global food import bill is almost twice as large as in normal times.
- The amplification is uneven across regions. Europe and Central Asia and the Middle East and North Africa experience the largest increases in food import bills, while all regions face systematically higher costs under high geopolitical risk.
- Heightened uncertainty, supply-chain frictions and macroeconomic pressures could amplify effects. During periods of heightened geopolitical tensions, higher precautionary demand, stronger cost pass-through, transport disruptions, higher insurance costs, and financial constraints magnify the transmission of oil supply shocks to food import bills.

Figure 1 shows the dynamic response of the global food import bill to an adverse oil supply shock that raises the oil price by 1 percent, under normal and high geopolitical risk conditions. Oil supply shocks consistently increase the food import bill at the global level, regardless of the geopolitical environment.

The analysis uses the impulse response function (IRF) estimated during a period of normal geopolitical conditions. This line, coloured green in the chart, shows the typical path the global food import bill follows after a shock when no unusual events occur. Deviations from this green line in the other scenario indicate how the same shock may play out differently under high level of geopolitical risk.

Under normal circumstances, the increase in the global food import bill is positive but relatively moderate, building gradually over the medium term. Specifically, FAO results show that an oil supply shock raising oil prices by 1 has a cumulative effect on the global food import bill of between 0.29 percent and 0.39 percent (midpoint 0.34 percent) after 10 months, before gradually reverting toward zero.

Figure 1. The dynamic impact of oil supply shocks on the global food import bill



Notes: The green (red) shaded area represents the 90 percent confidence intervals for the normal (high) level of geopolitical risk and are calculated using the Driscoll and Kraay standard errors. The impulse response function (IRF) is normalized such that the supply shock raises the oil price by 1 percent. The Geopolitical Risk (GPR) index used is developed by Dario Caldara and Matteo Iacoviello (2022). This index is a widely used measure in empirical macroeconomics and international finance. It captures the frequency of newspaper articles discussing geopolitical tensions, such as wars, terrorist attacks, and diplomatic conflicts. By interacting or conditioning the IRF on the GPR index, it allows to distinguish between oil supply shocks that arise from geopolitical events and those driven by other factors (e.g. demand shifts or cartel decisions). This distinction is critical because geopolitically driven oil shocks tend to have different transmission mechanisms, often generating greater uncertainty and more persistent effects on variables like the food import bill (FIB) compared to non-geopolitical oil shocks. *Source:* Authors' calculations.

When the same oil supply shock occurs during periods of high geopolitical risk (red impulse response function [IRF]) – based on the Geopolitical Risk (GPR) index developed by Caldara and Iacoviello (2022), which quantifies geopolitical tensions by measuring the frequency of adverse geopolitical events in leading international newspapers – the impact on the food import bill is substantially larger at all horizons. At the peak response, the increase in the food import bill under high geopolitical risk is almost twice as large as during normal periods (between 0.50 percent and 0.71 percent, with the midpoint at 0.61 percent).

For example, if supply disruptions to oil markets resulting from recent developments in the Middle East and the partial closure of the Strait of Hormuz were to push oil prices up by 10 percent, the global food import bill would be expected to increase between 5.0 percent and 7.1 percent within a few months.

This analysis indicates that geopolitical tensions serve as amplifiers, increasing the impact of oil disruptions on food markets. Heightened uncertainty, supply chain challenges, and increased risk premiums contribute to greater cost transmission, making oil supply interruptions, such as those in present circumstances, especially detrimental to food affordability, even when the magnitude of the oil shock remains constant.

Figure 2 demonstrates significant cross-regional variation in the effects of oil supply shocks on food import expenditures. Additionally, it confirms a consistent intensification of these impacts across all country groups during periods characterized by elevated geopolitical risk. For South-East Asia and the Pacific, the impact nearly doubles, increasing from between 0.15 percent and 0.44 percent (midpoint 0.30 percent) under normal conditions to 0.22 - 0.88 percent (midpoint 0.55 percent) during high-risk periods, reflecting the region's high exposure to imported energy and food commodities. Nevertheless, the associated confidence interval is wide, indicating considerable uncertainty around the magnitude of the response in the region.

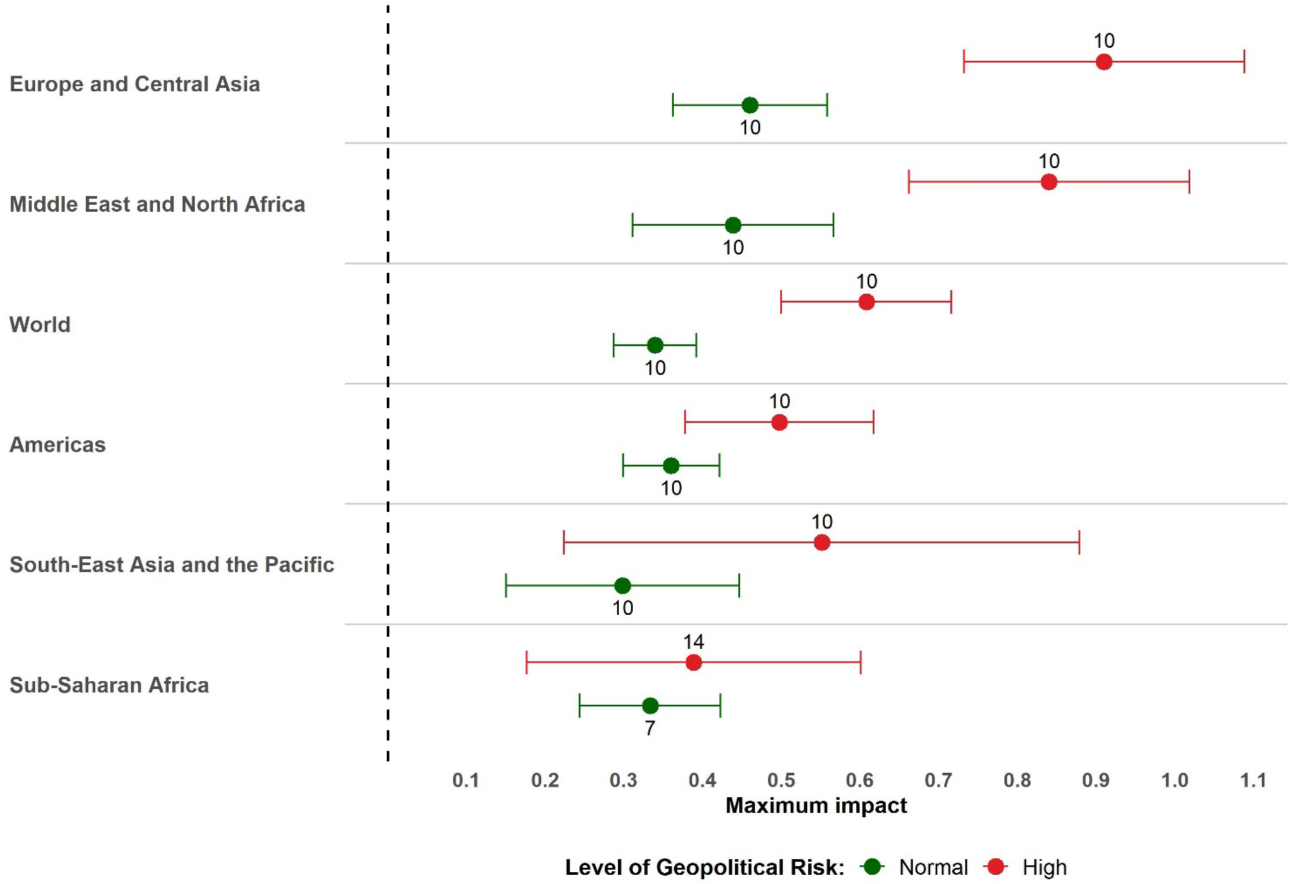
The amplification is particularly pronounced in Europe and Central Asia, where the food import bill response increases from between 0.36 percent and 0.56 percent (midpoint 0.46 percent) in normal times to the range 0.73 - 1.09 percent (midpoint 0.91 percent) during periods of elevated geopolitical risk, making this group one of the most affected in relative terms. In the Americas, oil supply shocks lead to a more moderate increase in the food import bill, rising from between 0.30 percent and 0.42 percent (midpoint 0.36 percent) in normal periods to 0.38 - 0.62 percent (midpoint 0.50 percent) during high geopolitical risk, suggesting comparatively lower, though still significant, vulnerability.

Middle East and North Africa exhibit one of the largest responses overall, with the food import bill increasing from between 0.31 percent and 0.57 percent (midpoint 0.44 percent) in normal times to 0.66 - 1.02 percent (midpoint 0.84 percent) under high geopolitical risk, consistent with strong exposure to regional tensions and high food import dependence despite energy exporter status.

For sub-Saharan Africa, the food import bill response rises from between 0.24 percent and 0.42 percent (midpoint 0.33 percent) in normal periods to 0.18 - 0.60 percent (midpoint 0.39 percent) during high geopolitical risk, indicating a proportionally smaller but more persistent impact, probably driven by the structural food import needs.

Overall, the regional evidence shows that while oil supply shocks increase food import bills everywhere, high geopolitical risk systematically magnifies their impact, with the largest absolute effects observed in Europe and Central Asia and the Middle East and North Africa, and sizeable relative increases across all regions.

Figure 2. The maximum impact of oil supply shocks on the food import bill, by region



Notes: The dots indicate the maximum cumulative impact of an oil supply shock on the food import bill. The capped spikes around the dot represent the 90 percent confidence interval calculated using Driscoll and Kraay standard errors. The number close to the dot indicates the month after the shock at which this peak is reached. *Sources:* Authors' own calculations.

Why do geopolitical tensions amplify the impact of the oil supply shocks on the FIB?

- **Higher uncertainty**. Periods of elevated geopolitical risk are associated with heightened uncertainty about future supply conditions, shipping routes, and policy responses. This raises precautionary demand for energy and food commodities and increases risk premia embedded in oil prices, making a given physical supply disruption translate into a larger and more persistent price increase.
- **Shipping insurance**. On 3 March, insurers expanded the list of dangerous zones in the Gulf. Two days later, protection and indemnity clubs - the insurers that cover ships - cancelled their war-risk policies for the region. As a result, insurance costs have skyrocketed. Before the crisis, a ship owner paid about 0.25 percent of the vessel's value to insure a voyage. Now, for high-risk ships, that cost can reach 10 percent. To make matters worse, insurance now only lasts for seven days at a time, meaning rates reset weekly. Insurers do not care about diplomatic progress. They care about

actual losses measured over quarters, not days. So even if hostilities cease tomorrow, normal shipping capacity will not return for months - because insurers will only lower premiums once they have seen a sustained period with no new claims.

- **Stronger cost pass-through along supply chains.** During geopolitical stress, firms face tighter margins, higher financing costs, and reduced ability to absorb input price shocks. As a result, increases in energy costs are passed through more quickly and more fully to food prices, magnifying the impact on food import bills.
- **Logistics and transport frictions.** Geopolitical tensions often disrupt maritime routes, raise insurance and freight costs, and reduce shipping capacity. Since food trade is highly transport-intensive, these frictions reinforce the transmission of oil price shocks to food import costs, especially for geographically distant import-dependent regions.
- **Weaker substitution and adjustment mechanisms.** In tranquil periods, countries and firms can partially adjust to oil shocks by switching suppliers, drawing on inventories, or smoothing imports over time. Under high geopolitical risk, these adjustment channels are impaired, leading to larger shorter impacts on import values.
- **Financial constraints and exchange rate pressures.** Elevated geopolitical risk is frequently associated with tighter global financial conditions and exchange rate volatility. For many emerging and developing economies, currency depreciation and higher financing costs further amplify the local-currency cost of food imports when oil prices rise.
- **Regional exposure and structural vulnerabilities.** The amplification is particularly strong in regions that combine high food import dependency with exposure to global energy market shocks. In these cases, geopolitical stress reinforces pre-existing vulnerabilities, leading to disproportionately large increases in food import bills.
- **Domestic policy responses.** Policy responses such as changes in fuel taxes, adjusting subsidies, changes in import/export procedures, or increasing insurance/security requirements tend to raise production and logistics costs in times of increased geopolitical tensions, further amplifying the effect of oil supply shocks on food import bills.

Methodological appendix

The dynamic effects of the oil supply shocks on the food import bill (FIB) are estimated by using panel local projections (Jordà, 2005; Jordà and Taylor, 2025). The specification is the following:

$$\Delta_h FIB_{i,t+h} = \alpha_{i,h} + \beta_h \varepsilon_t^{OS} + \sum_{j=1}^{l} \theta_j^h \Delta FIB_{i,t-j} + \sum_{j=1}^{l} \theta_j^h \Delta X_{i,t-j} + \rho t + \mu_{i,t+h}$$

where $\Delta_h FIB_{i,t+h}$ the time difference in the FIB between periods t – 1 and t + h. Expressed in logarithms, it captures the cumulative percentage change over that interval. $\alpha_{i,h}$ accounts for the country fixed effects; and $\Delta X_{i,t-j}$ the vector of lagged controls in first difference (world industrial production, WTI crude oil price, WB agricultural commodity price index, the global supply chain pressure index). ε_t^{OS} is an externally identified oil supply structural shock provided by Baumeister and Hamilton (2019), and β_h measures the cumulative impact of shipping costs on the FIB after h periods. We use 12 lags and all Vector Autoregressions (VARs) are deflated using the United States CPI.

To distinguish between normal and high geopolitical risk periods, we use the Geopolitical Risk (GPR) index developed by Caldara and Iacoviello (2022). The GPR is a news-based measure capturing the threat, realization, and escalation of adverse geopolitical events. It distinguishes between geopolitical threats and geopolitical acts (GPRA), with the latter focusing on the realization and intensification of conflicts, making it particularly relevant for assessing the macroeconomic impact of oil supply shocks during conflict episodes. To evaluate whether high geopolitical risk amplifies the food import bill response to an oil supply shock, we augment the baseline equation by interacting the shock with the GPR. Following Iacovella and Navarro (2019), we estimate the following:

$$\Delta_h FIB_{i,t+h} = \alpha_{i,h} + \beta_h \varepsilon_t^{OS} + \beta_h^{gpr} \left(e_{i,t-1}^{gpr} \varepsilon_t^{OS}\right)^{\perp} + \sum_{j=1}^{l} \theta_j^h \Delta FIB_{i,t-j} + \sum_{j=1}^{l} \theta_j^h \Delta X_{i,t-j} + \rho t + \mu_{i,t+h}$$

where $e_{i,t-1}^{gpr}$ is the exposure index to the geopolitical risk. The term $\left(e_{i,t-1}^{gpr} \varepsilon_t^{OS}\right)^{\perp}$ is obtained by orthogonalizing the interaction between the oil supply shocks and the logistic transformation of the standardized GPR, re-centered between its 50th and 90th percentile. By doing so, β_h is the response to the shock when GRP is at their median value, and $\beta_h + \beta_h^{gpr}$ is the response when the exposure is at the 90th percentile of its distribution. Instead, orthogonalization ensures that β_h remain the same as in the previous equation while β_h^{gpr} is the marginal effect of the geopolitical risk on the response of the FIB to an oil supply shock when GRP moves from its 50th percentile (normal times) to its 90th percentile (high turmoil with adverse geopolitical events). Data sample covers the period JAN2005 - DEC2022 and almost 200 countries and regions.

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